

Ansh Pethani

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EDUCATION

North Carolina State University, Raleigh, NC, USA
Master of Computer Science
Vellore Institute of Technology, Vellore, IN
Bachelor of Technology in Information Technology

Aug 2025 - May 2027
GPA: 4/4
Sep 2021 - May 2025
CGPA: 8.91/10

Publications

- Pethani, A., Nallakaruppan, M.K. (2024). Electronic Copyright and Legal Application Regarding Non-Fungible Tokens (NFTs). In: Bhattacharyya, S., Das, G., De, S., Mrcic, L. (eds) Recent Trends in Intelligence Enabled Research. DoSIER 2023. Advances in Intelligent Systems and Computing, vol 1457. Springer, Singapore. [\[Link\]](#)
- Nallakaruppan, M.K., Pethani, A. and Pelusi, D. (2025), "Data Security in the Age of Marketing: Safeguarding Customer Information and Compliance", Balusamy, B., Grover, V., Nallakaruppan, M.K., Rajasekaran, V.A. and Milanova, M. (Ed.) Data Engineering for Data-driven Marketing, Emerald Publishing Limited, Leeds, pp. 165-178.[\[Link\]](#)

PERSONAL EXPERIENCE

Research Associate, iEXCEL Lab, CS Department, NC State University, USA

Aug 2026 - Present

- Managing CI/CD pipelines and backend deployments, maintaining backend integrations, and adding additional features to webpages.
- Engineering a GNN to validate trophic interactions and generate interaction plausibility scores from crowdsourced food web data.
- Integrating pre-trained multi-species computer vision pipelines into existing backend and frontend architectures.
- Facilitating expert validation studies for a vision-model-based alt-text generator and deploying analytical tools driven by large-scale publication datasets.

Research Associate, IEC Lab, CS Department, NC State University, USA

Aug 2025 - Aug 2026

- Developed Python ETL pipelines to parse, clean, and resolve various conflicts across CSV, XML, & JSON records.
- Developed and optimized an LLM-based Python pipeline for automated quiz generation using student responses.
- Researched embedding models and deployed NV-Embed-v2 to generate embeddings, evaluating them using K-Means clustering.

AI Development Intern, IoTIoT, India

Jan 2025 - Jul 2025

- Developed AI/ML systems, including an offline chatbot (Ollama, Flask, Tauri, Rust), image detection models using TensorFlow/Keras (from scratch and with transfer learning), and a zero-shot voice cloning model.
- Engineered generative AI applications using Gemini/OpenAI APIs, built a Google A2A wrapper for WhatsApp MCP, and documented technical learnings through AI-focused blogs.

Python Backend and AI Development Intern, DeepKlarity, India

Sep 2024 - May 2025

- Engineered ML & AI solutions, including YOLOv11-based image detection models (dog-nose & cat-face) and generative AI-powered chatbots (blog generation, RAGs for document/PDF/image Q&A).
- Built and deployed applications across domains: Gmail add-on for automated email risk assessment, AutoEDA tool for PDF-based data analysis, and a data analytics platform using Python and generative AI.
- Designed and built full-stack web applications using React, JavaScript, FastAPI, and PostgreSQL, developing REST APIs and AI-powered features that improved automation, scalability, and client usability.

KEY PROJECTS

- **NBA/ABA Statistics and Future Predictions (2025)**: Performed data mining, association mining, and applied ML models to analyze and predict game and player performance outcomes. Tech used: Python, Scikit-learn, Pandas, Numpy
- **IDS Attack Detection in IoV CAN Bus (2024)**: Evaluated 12 ML/DL models on a 1.4M-instance dataset, applied SMOTE, and optimized accuracy/efficiency. Tech Used: Python, Tensorflow, Scikit-learn, Imbalance-learn, Pandas, NumPy
- **Real-Time Analysis (2024)**: Developed a data processing app to analyze web-scraped data at 5-second intervals, featuring automatic graph visualizations as new data batches were received; Tech Used: Python, PySpark, Hadoop
- **Socket Programming Chat App (2023)**: Developed an application in Python using sockets, enabling communication between multiple systems on the same network via TCP protocol.

ADDITIONAL INFORMATION

- **Languages**: Python, JavaScript, Java, c, C++, SQL
- **AI/ML**: OpenAI API, Gemini API, Ollama, TensorFlow, Scikit-Learn, Keras, Numpy, Pandas
- **Frontend**: React, HTML, CSS | **Backend**: FastAPI, Flask, REST APIs | **Databases**: PostgreSQL, ChromaDB
- **Tools**: Git, Linux, Postman, VS Code, Rust, Tauri
- **Coursework**: Design and Analysis Of Algorithms, Automated Learning & Data Analysis, SE, AI, ML, DSA, Computer & Network Security, Cloud Computing, Big Data, Soft Computing, Computer Networks, DBMS, Operating Systems, Information Security, Cyber Security, Computer Architecture, Network Information Security, Advanced Robotics, Parallel Systems